

Assignment 2: Weekly Progress Report

Name: Manoj A B

Reg No: 24UG00520

Week 1 Report

Activities Completed

- Conducted data sourcing for FIFA World Cup statistics, including team performance, player stats, and match results using online tools and APIs.
- Developed a custom web scraper to extract relevant data from a new sports-related website not covered by existing tools, and documented the scraping code, targeted site, collected data fields, scraping logic, and challenges faced.
- Performed comprehensive data cleaning: removing duplicates, handling missing values by imputation or removal, and preparing the dataset for modeling.
- Engineered features relevant to the task, including team average age, FIFA ranking, goal differences, player experience, and win rate. Documented the rationale for each feature.

Outcomes

- Produced a cleaned dataset along with a data description report detailing sources, cleaning process, feature explanation, and scraper documentation including code snippets and instructions.
- Ensured all steps were well-documented for transparency and reproducibility.

Week 2 Report

Activities Completed

- Started model development: implemented at least two classification models (such as logistic regression and random forest) to predict World Cup finalists.
- Described preprocessing steps for model building, which included data scaling, encoding categorical variables, feature selection, and hyperparameter tuning methods.
- Conducted train-test split or k-fold cross-validation to evaluate models fairly and prevent overfitting.
- Provided fully commented code notebooks/scripts documenting the entire process and choices made during model implementation.

Outcomes

- Developed and documented complete code for all models, ensuring clear explanations of methodology and approach.
- Summarized model training, tuning, and validation procedures in the report, laying the groundwork for next week's evaluation and interpretation.